DeepFER: Facial Emotion Recognition using Deep Learning

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Github link - https://github.com/prem1555/DeepFER-Deep-Facial-Emotion-Recognition

The objective of this project was to develop an automated Facial Emotion Recognition (FER) system capable of classifying facial images into seven emotional categories: Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise. The project employed Deep Learning techniques including a Custom Convolutional Neural Network (CNN) and a Transfer Learning model based on MobileNetV2

The dataset consisted of RGB facial images of size 48×48 pixels. Comprehensive preprocessing steps such as label encoding, one-hot encoding, normalization, train validation-test splitting, and data augmentation were applied. Two models were developed and compared. The Custom CNN achieved a test accuracy of approximately 38.61%, while MobileNetV2 achieved a higher test accuracy of approximately 47.63%.

Comparative evaluation demonstrated that MobileNetV2 provided superior feature extraction capabilities and better generalization performance. Consequently, MobileNetV2 was selected as the final model and deployed using Streamlit to provide real-time emotion prediction from uploaded facial images.

TABLE OF CONTENTS

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1. INTRODUCTION

Emotion recognition is an important field within Artificial Intelligence and Computer Vision. Human emotions influence communication, decision-making, and behavior. Enabling machines to understand emotions can significantly improve interactions between humans and intelligent systems.

Traditional approaches relied on handcrafted features and rule-based systems. However, these approaches often struggled to generalize across different individuals and environments. Deep Learning techniques, particularly Convolutional Neural Networks (CNNs), have revolutionized image analysis by automatically learning hierarchical feature representations.

This project explores the application of CNNs and Transfer Learning for automated facial emotion recognition.

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2. PROBLEM STATEMENT

The goal of this project is to develop a robust and efficient system capable of accurately classifying facial expressions into seven emotional categories using deep learning techniques.

The system should:

• Process facial images automatically.

• Recognize seven emotion classes.

• Generalize to unseen facial images.

• Be deployable through a user-friendly interface.

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3. PROJECT OBJECTIVES

Primary Objectives:

- Develop a deep learning-based emotion recognition system.
- Compare CNN and Transfer Learning approaches.
- Evaluate model performance using multiple metrics.
- Deploy the final model through a Streamlit application.

Specific Objectives:

- Perform dataset exploration and preprocessing.
- Apply data augmentation techniques.
- Train and evaluate multiple deep learning models.
- Select the best-performing model.
- Enable real-time emotion prediction.

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4. DATASET DESCRIPTION

Dataset Characteristics:

- Image Type: RGB Images
- Resolution: 48 × 48 Pixels
- Number of Classes: 7

Emotion Categories:

1. Angry

2. Disgust

3. Fear

4. Happy

5. Neutral

6. Sad

7. Surprise Dataset Integrity:

• No corrupted images detected.

• Consistent image dimensions across all samples.

• Balanced preprocessing pipeline established. [Insert Dataset Distribution Screenshot]

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5. DATA UNDERSTANDING AND EXPLORATION

Class Distribution Analysis Discuss:

• Distribution of images across classes.

• Potential class imbalance.

• Impact on model learning.

Visual Explorations:

• Emotion Distribution Bar Chart

• Percentage Distribution Pie Chart

• Sample Images from Each Class

• Random Image Visualization

• Pixel Intensity Distribution

Insights:

• Happy and Neutral classes contain more samples.

• Disgust class contains fewer samples.

• Dataset contains significant variability in facial appearance.

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6. DATA WRANGLING

Steps Performed:

• Loaded images from class folders.

• Converted images into NumPy arrays.

• Created label arrays.

• Verified image dimensions.

• Verified class labels.

• Created final feature and target datasets.

Outcome:

• Feature Dataset (X)

• Target Dataset (y)

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7. DATA PREPROCESSING

Preprocessing Steps:

1. Label Encoding

2. One-Hot Encoding

3. Train-Test Split

4. Train Validation Split

5. Pixel Normalization

6. Data Augmentation

Data Augmentation Techniques:

- Rotation
- Horizontal Flip
- Width Shift
- Height Shift
- Zoom

Benefits:

- Reduced overfitting
- Improved generalization
- Increased training variability

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8. MODEL DEVELOPMENT

Model 1: Custom CNN

Architecture:

- Conv2D

• Batch Normalization

• Max Pooling

• Dropout

• Dense Layers

• SoftMax Output

Why CNN?

CNNs automatically learn facial features and spatial relationships from image data.

Training Configuration:

• Optimizer: Adam

• Loss Function: Categorical Crossentropy

• Metrics: Accuracy

Model 2: MobileNetV2 Transfer Learning

Why MobileNetV2?

• Lightweight architecture

• Pretrained on ImageNet

• Faster convergence

• Better feature extraction

Architecture:

• MobileNetV2 Base Model

• Global Average Pooling

• Dense Layers

• Softmax Output Layer

9. MODEL EVALUATION

Evaluation Metrics:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

Model 1 Results

Test Accuracy: 38.61%

Observations:

- Strong performance on Happy emotion.
- Poor recognition of Angry, Fear, Disgust, and Surprise.

Model 2 Results

Test Accuracy: 47.63% Observations:

- Better class-wise performance.
- Improved recognition of multiple emotions.

10. MODEL COMPARISON

Metric	Custom CNN	MobileNetV2
Accuracy	38.61%	47.63%
Weighted F1 Score	0.31	0.46

Discussion: MobileNetV2 consistently outperformed the Custom CNN due to pretrained feature representations.

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11. FINAL MODEL SELECTION

MobileNetV2 was selected because:

- Highest accuracy.
- Better generalization.
- Improved class-wise performance.
- Efficient deployment suitability.

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12. DEPLOYMENT

Deployment Framework:

- Streamlit Deployment Features:
- Image Upload
- Automatic Preprocessing
- Real-Time Emotion Prediction
- Confidence Score Display Application Workflow:

User Upload → Preprocessing → Model Prediction → Emotion Output

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13. RESULTS AND FINDINGS

Key Findings:

- Transfer Learning significantly improved performance.

- MobileNetV2 generalized better than CNN.
- Data augmentation improved robustness.
- Emotion recognition remains challenging due to similarities between emotions.

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14. CHALLENGES FACED

- Limited image resolution (48×48).
- Similar facial expressions among emotions.
- Class imbalance.
- Computational constraints during CNN training.
- Difficulty distinguishing Fear and Surprise.

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15. FUTURE SCOPE

- Webcam-based real-time emotion recognition.
- Multi-face detection.
- Fine-tuned transfer learning models.
- Cloud deployment.
- Mobile applications.
- Larger and higher-quality datasets.

16. CONCLUSION

The project successfully developed a deep learning-based facial emotion recognition system. Two models were implemented and evaluated. The MobileNetV2 transfer learning model achieved the best performance with a test accuracy of 47.63% and was selected for deployment. The project demonstrates the effectiveness of transfer learning in computer vision tasks and provides a strong foundation for future emotion-aware intelligent systems.